

GRASHOF'S LAW

$$S + L \text{ vs } P + Q$$

GRASHOF

$$S + L < P + Q$$

NON-GRASHOF

$$S + L > P + Q$$

Crank-Rocker

Shortest = input

Input: full rotation

Output: oscillates

Double-Crank

Shortest = ground

Input: full rotation

Output: full rotation

Rocker-Crank

Shortest = output

Input: oscillates

Output: full rotation

Triple-Rocker

No link can fully rotate

All links oscillate

(Limited motion range)

SPECIAL CASE: $S + L = P + Q$

Change-point mechanism (passes through dead points)